

● MEDIUM

● ALGEBRA

Algebra

10 questions · ~15 min

03

Q1

GRID-IN ALGEBRA

A line in the xy -plane has a slope of 9 and passes through the point $(0, -5)$. The equation $y = px + r$ defines the line, where p and r are constants. What is the value of p ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A bowl contains 20 ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by 1 ounce every 4 days. If 9 ounces of water remain in this bowl, for how many days has it been uncovered?

- A. 3
- B. 7
- C. 36
- D. 44

For the linear function h , the graph of $y = hx$ in the xy -plane passes through the points $(7, 21)$ and $(9, 25)$. Which equation defines h ?

A. $hx = \frac{1}{2}x - \frac{7}{2}$

B. $hx = 2x + 7$

C. $hx = 7x + 21$

D. $hx = 9x + 25$

Q4

GRID-IN

ALGEBRA

$$y = -2x$$

$$3x + y = 40$$

The solution to the given system of equations is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

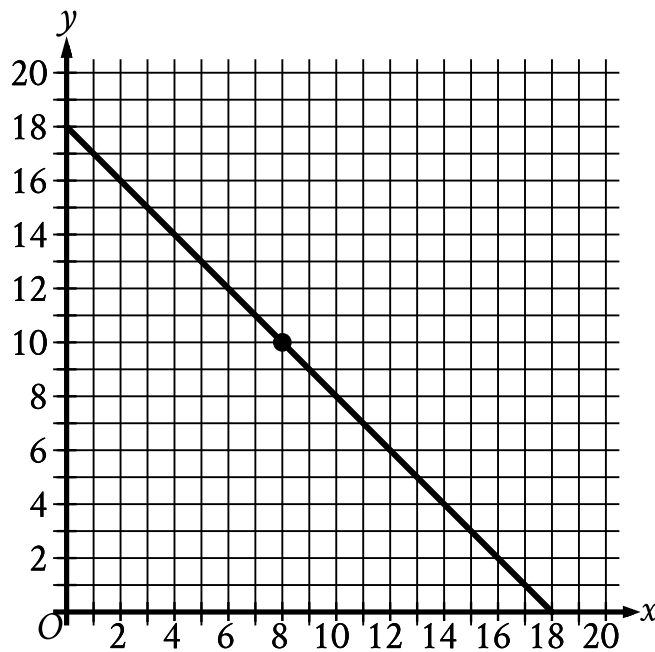
The average annual energy cost for a certain home is \$4,334. The homeowner plans to spend \$25,000 to install a geothermal heating system. The homeowner estimates that the average annual energy cost will then be \$2,712. Which of the following inequalities can be solved to find t , the number of years after installation at which the total amount of energy cost savings will exceed the installation cost?

A. $25,000 > (4,334 - 2,712)t$

B. $25,000 < (4,334 - 2,712)t$

C. $25,000 - 4,334 > 2,712t$

D. $25,000 > \frac{4,332}{2,712}t$



- The line slants gradually down from left to right.
- The line passes through the following points:
 - (8 comma 10)
 - (9 comma 9)

The graph in the xy -plane models the possible combinations of length x , in meters (m), and width y , in meters, for a rectangle with a perimeter of 36 m. Which statement is the best interpretation of the point 8, 10 in this context?

- A. The length is 10 m less than the perimeter, and the width is 8 m less than the perimeter.
- B. The length is 10 m, and the width is 8 m.
- C. The length is 8 m, and the width is 10 m.

- D. The length is 8 m less than the perimeter, and the width is 10 m less than the perimeter.

If $2(x-5)+3(x-5)=10$, what is the value of $x-5$?

- A. 2
- B. 5
- C. 7
- D. 12

A linear function f gives a company's profit, in dollars, for selling x items. The company's profit is \$ 220 when it sells 8 items, and its profit is \$ 320 when it sells 10 items. Which equation defines f ?

A. $fx = 150x - 320$

B. $fx = 32x$

C. $fx = 50x - 10x$

D. $fx = 50x - 180$

Q9

GRID-IN

ALGEBRA

$$x + 3y = 29$$

$$3y = 11$$

The solution to the given system of equations is x, y . What is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The minimum value of x is 12 less than 6 times another number n . Which inequality shows the possible values of x ?

A. $x \leq 6n - 12$

B. $x \geq 6n - 12$

C. $x \leq 12 - 6n$

D. $x \geq 12 - 6n$